

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering **ASSESSMENT TOOL 3 – Comparative Analysis**

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO4 – Articulation (combining methods for evaluation) P4	Assessment:	Assessment Tool – Comparative Analysis
Weightage:	10% of CIA	Total Marks:	100
CO4 Statement:	<i>Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)</i>		
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Section / Batch:	2024 AIML	Date:	23/8/2026

Code of Conduct

I M.MANASA(192425189)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: m.manasa_____

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Comparative Analysis

10. Error Rate Comparison

Given Data

Method	Errors	Total
A	20	100
B	10	100

(a) Computation of Error Rate

Error Rate = (Number of Errors / Total Instances) $\times$ 100

Method A: Error Rate = (20 / 100) $\times$ 100 = 20%

Method B: Error Rate = (10 / 100) $\times$ 100 = 10%

Comparison Table

Method	Error Rate (%)	Accuracy (%)
A	20%	80%
B	10%	90%

(b) Comparison and Evaluation

Method B has an error rate of 10%, which is exactly half the error rate of Method A (20%). This means Method B correctly recognizes/classifies a higher proportion of instances (90% accuracy) compared to Method A (80% accuracy).

Evaluation: Method B is the more reliable and accurate technique. A lower error rate indicates that Method B's model or algorithm generalizes better to the given dataset and produces fewer misclassifications, which is critical in vision tasks such as object and pattern recognition where false detections can propagate errors downstream (e.g., in tracking or scene understanding). Therefore, Method B is preferred over Method A for this evaluation criterion.

11. Motion Estimation Variance

Given Data

Frame	A	B
1	2	6
2	3	7
3	2	5

(a) Computation of Mean Error

Mean Error = (Sum of Errors) / (Number of Frames)

Method A: Mean = (2 + 3 + 2) / 3 = 7 / 3 = 2.33

Method B: Mean = (6 + 7 + 5) / 3 = 18 / 3 = 6.00

Computation of Variability (Variance & Standard Deviation)

Variance = $\Sigma(x_i - \text{mean})^2 / n$

Method A: Deviations from mean (2.33): -0.33, 0.67, -0.33 $\rightarrow$ Squared: 0.11, 0.44, 0.11 $\rightarrow$ Sum = 0.66

Variance(A) = $0.66 / 3 = 0.22$ | Standard Deviation(A) = $\sqrt{0.22} \approx 0.47$

Method B: Deviations from mean (6.00): 0, 1, -1 → Squared: 0, 1, 1 → Sum = 2.00

Variance(B) = $2.00 / 3 = 0.67$ | Standard Deviation(B) = $\sqrt{0.67} \approx 0.82$

Comparison Table

Method	Mean Error	Variance	Std. Deviation
A	2.33	0.22	0.47
B	6.00	0.67	0.82

(b) Comparison of Variability and Stability Evaluation

Method A has both a lower mean error (2.33 vs 6.00) and lower variance (0.22 vs 0.67) than Method B. This means Method A not only makes fewer motion-estimation errors on average, but also produces errors that are more consistent (less spread out) across frames.

Evaluation: Method B shows higher variability (larger standard deviation), which indicates unstable or inconsistent motion estimation — its performance fluctuates more across frames. Method A demonstrates greater stability and consistency, making it the more dependable technique for motion estimation in sequential video frames, where consistent tracking accuracy is essential for reliable motion analysis.

12. IoU (Intersection over Union) Comparison

Given Data

Object	A	B
1	0.60	0.80
2	0.70	0.85
3	0.65	0.90

(a) Computation of Average IoU

Average IoU = (Sum of IoU values) / (Number of Objects)

Method A: Average IoU = $(0.60 + 0.70 + 0.65) / 3 = 1.95 / 3 = 0.65$

Method B: Average IoU = $(0.80 + 0.85 + 0.90) / 3 = 2.55 / 3 = 0.85$

Comparison Table

Method	Average IoU	Detection Quality
A	0.65	Moderate overlap
B	0.85	High overlap

(b) Comparison and Evaluation of Detection Quality

Method B achieves a higher average IoU (0.85) compared to Method A (0.65). Since IoU measures the overlap between the predicted bounding box and the ground-truth bounding box (values range from 0 to 1, with 1 being a perfect match), a higher IoU directly indicates more accurate object localization.

Evaluation: Using the common benchmark threshold of $\text{IoU} \geq 0.5$ for a valid detection, both methods qualify, but Method B (0.85) falls well within the 'high-quality detection' range typically used in object detection literature (e.g., strict evaluation at $\text{IoU} \geq 0.75$), whereas Method A (0.65) only meets a moderate-quality threshold. Therefore, Method B provides superior bounding-box localization and overall detection quality compared to Method A.

Overall Summary

Question	Better Method	Key Metric
Q10 – Error Rate	B	10% vs 20%
Q11 – Motion Estimation	A	Mean 2.33 & Var 0.22 vs Mean 6.00 & Var 0.67
Q12 – IoU	B	0.85 vs 0.65